

MULTIVARIATE EXPONENTIAL DISCRETE DISTRIBUTIONS AND THEIR CHARACTERIZATION BY THE RAO-RUBIN CONDITION FOR THE ADDITIVE DAMAGE MODEL

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Summary: The well-known additive damage model introduced by Rao and Rubin (1964) has been extended to the multivariate situation along with the Rao-Rubin (RR) condition. This paper presents some multivariate exponential discrete distributions which include the Lagrangian Poisson and the multivariate Lagrangian negative binomial distributions along with the multivariate Lagrangian negative binomial which is characterized by using the survival distribution as the multivariate generalized Markov-Polya and the multivariate RR condition. Conversely, the survival distribution is characterized using the RR condition and the multivariate Lagrangian distributions for the original observations. Many well-known results can be derived as special cases.

1. INTRODUCTION

A random vector variable with p components will be denoted by $Z' = (Z_1, \dots, Z_p)$. By some chance mechanism, Z will be

reduced to the vector X , and then $Y = Z - X$ will be called the damaged part of Z . When Z is undamaged, $Z = X$ or $Y = 0$. We may observe that some components of Z may be damaged while some other components may be undamaged. All such situations will constitute the damaged Z . The reduction to X is defined by the conditional distribution $s(x|z) = P(X = x|Z = z)$ and this will be called the survival distribution. The distribution of the original vector variable Z will be denoted by $P(Z = z) = h(z)$. Thus, the damaged model will be characterized by the survival distribution, $s(x|z)$, and the distribution of Z , $h(z)$. The problem is to determine $h(z)$ or $s(x|z)$.

1.A Multivariate distributions

Multivariate discrete exponential family of distributions was studied in some detail by Khatri (1983 b) and this family is given by

$$(1.1) \quad P(X=x) = a(x)\exp(\phi'x - Q(\theta)) \quad \text{for } x \in T$$

where $\phi' = (\phi_1, \phi_2, \dots, \phi_p)$, $\phi_i = R_i(\theta)$, $i=1,2,\dots,p$ and

$$(1.2) \quad \exp(Q(\theta)) = \sum_{x \in T} a(x)\exp(\phi'x).$$

Here, the sample space T is given by $\{x: a(x) \geq 0\}$, but generally $T = \{x: x_i = 0, 1, \dots \text{ for all } i=1, 2, \dots, p\}$. For the probability function (1.1), $a(x) \geq 0$ for all $x \in T$. θ is known as the parameter of the distribution and (1.1) will be valid for $\theta \in \mathcal{D}$, known as the parametric space. Khatri (1983 a,b) has obtained various properties of this distribution including some important special cases like the multivariate Lagrangian Poisson, the Lagrangian multinomial, the multivariate Lagrangian logarithmic and the

multivariate Lagrangian negative binomial. For the derivation of such special cases, one can refer to Khatri (1983a).

If $Q(\theta)$ and ϕ are infinitely differentiable with respect to θ at $\theta = 0 \in \mathcal{D}$, then $a(x)$ of (1.2) can be obtained by using the method of Good (1960) provided that the limits of the functions

$$(1.3) \quad f_i(\theta) = \theta_i \exp(-\phi_i) \quad \text{for } i = 1, 2, \dots, p$$

as $\theta \rightarrow 0$ are nonzero for all i . Under these assumptions, $a(x)$ can be calculated from

$$(1.4) \quad (x!)a(x) = [D^x \{ \exp(Q(\theta)) (f(\theta))^x | I - A(\theta) | \}]_{\theta=0}$$

where $D_i = \partial / \partial \theta_i$,

$$(1.5) \quad D^x = \prod_{i=1}^p D_i^{x_i}, \quad x! = \prod_{i=1}^p (x_i!), \quad (f(\theta))^x = \prod_{i=1}^p (f_i(\theta))^{x_i}$$

and

$$(1.6) \quad A(\theta) = (a_{ij}(\theta)), \quad a_{ij}(\theta) = \theta_i D_j \ln f_i(\theta) \quad \text{for all } i, j.$$

Here, the sample space T is taken as $T: \{x: x_i = 0, 1, \dots \text{ for all } i = 1, 2, \dots, p\}$. The main difficulty is to find the constant $a(x)$ explicitly for given $Q(\theta)$ and ϕ .

Let $Q(\theta) = \sum_{i=1}^p m_i \theta_i = m' \theta$ and let $a(x)$ of (1.4)

be denoted by $a(x, m)$ to emphasize the parameter m . Let $m \in \mathcal{D}_1$ be the parametric space of m such that $a(x, m) \geq 0$ for all $x \in T$ and $m \in \mathcal{D}_1$. The distribution (1.1) in this situation will be rewritten as

$$(1.7) \quad P(X=x) = a(x, m) \{ \theta^x / (f(\theta))^x \} \exp(-m' \theta) \quad \text{for all } x \in T,$$

where $a(x, m)$ can be determined from (1.4) with $Q(\theta) = m'\theta$, $\theta \in \mathcal{D}$ and $m \in \mathcal{D}_1$.

Now, suppose that X and Y are independently distributed and their respective distributions are given by (1.7) with $m \in \mathcal{D}_1$ and $n \in \mathcal{D}_1$. Then, the distribution of $Z = X + Y$ is the same as (1.7) with $m + n \in \mathcal{D}_1$. Hence, in this situation, the conditional distribution of X , given Z is given by

$$(1.8) \quad s(x|z) = P(X=x|Z=z) = \frac{a(x, m)a(z-x, n)}{a(z, m+n)}$$

for all $x \in T_1(z) : \{x : x_i = 0, 1, 2, \dots, z_i \text{ for all } i \text{ and } m, n \text{ and } m+n \text{ all in } \mathcal{D}_1\}$. This survival distribution is the generalization of many survival distributions considered by various authors like Consul (1975), Patil and Ratnaparkhi (1975) and Janardan (1975 a).

Now, we shall give a generalized version of the Markov-Polya distribution for the univariate situation. The multivariate Markov-Polya distribution is given after this univariate distribution. For the univariate case, define

$$(1.9) \quad f(\theta) = c\theta(1 - \exp(-c\theta))^{-1}(1 + d\theta)^b,$$

and use (1.4) for calculating $a(x, m)$. Here

$$\begin{aligned} (1.10) \quad x!a(x, m) &= [D^x \{ \exp(m\theta) (f(\theta))^x (1 - \theta D \ln f(\theta)) \}]_{\theta=0} \\ &= m [D^{x-1} \exp(m\theta) (f(\theta))^x]_{\theta=0} \\ &= m \left[D^{x-1} \left\{ \left(\frac{c\theta}{1 - \exp(-c\theta)} \right)^x \exp(m\theta) (1 + d\theta)^{bx} \right\} \right]_{\theta=0} \\ &= m \sum_{j=0}^{x-1} \binom{x-1}{j} \left[D^{x-1-j} \left\{ \left(\frac{c\theta}{1 - \exp(-c\theta)} \right)^x \exp(m\theta) \right\} \right]_{\theta=0} \\ &\quad \left[D^j (1 + d\theta)^{bx} \right]_{\theta=0} \end{aligned}$$

$$= m \sum_{j=0}^{x-1} \binom{x-1}{j} \binom{bx}{j} (j!) d^j (1+mc)^{(x-1-j, c)}$$

where $(m+c)(0, c) = 1$, $(m+c)(y, c) = (m+c)(m+2c)\dots(m+yc)$.

Observe that if $t = bd$ is fixed, then after taking limits as $b \rightarrow \infty$ (and $d \rightarrow 0$), we get

$$(1.11) \quad x!a(x, m) = m \sum_{j=0}^{x-1} \binom{x-1}{j} (tx)^j (m+c)^{(x-1-j, c)} = J_x(m, c, t)$$

where

$$(1.12) \quad J_x(m, c, t) = \frac{m}{m+xt} (m+xt)^{(x, c)}$$

Thus, (1.12) is a particular case of (1.10). The univariate (1.7) distribution for (1.10) is given by

$$(1.13) \quad P(X=x) = a(x, m) \{1 - \exp(-c\theta)/c\} (1+d\theta)^{-bx} \exp(-m\theta)$$

and as $b \rightarrow \infty$ (or $d \rightarrow 0$) and $bd = t$ (fixed), (1.13) tends to

$$(1.14) \quad P(X=x) = \frac{1}{x!} J_x(m, c, t) v^x (1-cv)^{(xt+m)/c}$$

with $v = (1 - \exp(-c\theta))/c$. The parameters m , c and t are related to each other so that $J_x(m, c, t) \geq 0$ for all $x \in T$.

The parameters m , c , b and d in (1.13) have similar restrictions. The survival distribution based on the distribution (1.13) is given by

$$(1.15) \quad s(x|z) = P(X=x|Z=x) = \frac{a(x, m)a(z-x, n)}{a(z, m+n)}$$

where $a(x, m)$ is defined by (1.10). This is a generalization of the Markov-Polya distribution which can be obtained from (1.15) as a limit $b \rightarrow \infty$ such that $bd = t$ fixed.

er, if $c=1$, $m > 0$ and $t > 0$ in (1.14), then it is as the generalized negative binomial (or Lagrangian negative binomial distribution) as given by Jain and Consul (1). In this situation, the survival distribution (1.8) given by

$$5) \quad s(x|z) = P(X=x|Z=z) = \binom{z}{x} \frac{J_x(m, c, t) J_{z-x}(n, c, t)}{J_z(m+n, c, t)}$$

all $x = 0, 1, \dots, z$. This distribution is known as the bivariate Polya distribution which includes many known distributions like the hypergeometric distribution, the Lagrangian binomial etc. as special cases. Consul and Mittal (1975) and Erdan (1975 b, 1978) arrived at the above distribution in study of urn problems with predetermined strategies.

can generalize the above univariate distribution to the multivariate situation if we choose

$$7) \quad f_i(\theta) = c_i \theta_i (1 - \exp(-c_i \theta_i))^{-1} \exp\left(\sum_{j=1}^p d_{ij} \theta_j\right) \\ i = 1, 2, 3, \dots, p.$$

Chen (1983 a) chose $f_i(\theta)$ as $\lim_{c_i \rightarrow 0} f_i(\theta)$ and obtained the

multivariate Lagrangian Poisson. Thus, (1.17) will generalize the multivariate Lagrangian Poisson distribution. To obtain explicit expression of $a(x, m)$, we have to use (1.4). In this case

$$3) \quad (x!)a(x, m) = [D^x \{ \exp(y' \theta) (g(\theta))^x | B(\theta) \}]_{\theta=0}$$

$$1) \quad (g(\theta))^x = \prod_{i=1}^p (g_i(\theta_i))^{x_i}, \quad g_i(\theta_i) = c_i \theta_i (1 - \exp(-c_i \theta_i))^{-1}$$

$$(1.20) \quad D = (d_{ij}), \quad y = D'x + m, \quad B(\theta) = D_g - D_g(D + D_c)$$

with $D_\theta = \text{diag}(\theta_1, \theta_2, \dots, \theta_p)$,

$D_g = \text{diag}(g_1(\theta_1), g_2(\theta_2), \dots, g_p(\theta_p))$, and

$D_c = \text{diag}(c_1, c_2, \dots, c_p)$. Let us define

$$(1.21) \quad h(x_\alpha, y_\alpha, c_\alpha) = D_\alpha^{\alpha} \left\{ \exp(y_\alpha \theta_\alpha) (g_\alpha(\theta_\alpha))^{x_\alpha + 1} \right\}_{\theta_\alpha=0} \\ = (c_\alpha + y_\alpha)^{(x_\alpha, c_\alpha)} \\ = (y_\alpha + c_\alpha)(y_\alpha + 2c_\alpha) \dots (y_\alpha + x_\alpha c_\alpha).$$

Now, expand $|B(\theta)|$ by the diagonal elements and we can write

$$(1.22) \quad |B(\theta)| = \sum_{k=1}^p \sum_{j \in P_k} e(j) \prod_{\alpha=1}^p \{g_{j_\alpha}(\theta_{j_\alpha})\} \prod_{\beta=k+1}^p (\theta_{j_\beta})$$

where $j = (j_1, j_2, \dots, j_p)$ is a permutation of $(1, 2, \dots, p)$

such that for a given k , $1 \leq j_1 < j_2 < \dots < j_k \leq p$,

$1 \leq j_{k+1} < j_{k+2} < \dots < j_p \leq p$ and P_k is a class of such

permutations. Here $e(j)$ depends on the elements of $D + D_c$ and it is a determinant of a sub-matrix obtained from $-(D + D_c)$

after deleting the $(j_1\text{-th}, \dots, j_k\text{-th})$ rows and the $(j_1\text{-th}, \dots, j_k\text{-th})$ columns. Now, it is easy to see that (1.14) is equivalent to

$$(1.23) \quad (x!)a(x, m) = \sum_{k=1}^p \sum_{j \in P_k} e(j) \left[\prod_{\alpha=1}^p D_{j_\alpha}^{x_{j_\alpha}} \left\{ \exp(y_{j_\alpha} \theta_{j_\alpha}) (g_{j_\alpha}(\theta_{j_\alpha}))^{x_{j_\alpha} + 1} \right\} \right]_{\theta=0} \\ \times \left[\prod_{\beta=k+1}^p D_{j_\beta}^{x_{j_\beta}} \left\{ \exp(y_{j_\beta} \theta_{j_\beta}) (g_{j_\beta}(\theta_{j_\beta}))^{x_{j_\beta} + 1} \right\} \right]_{\theta=0}$$

and using (1.21) in (1.23), we get

$$(1.24) \quad (x!)a(x, m) = \sum_{k=1}^p \sum_{j \in P_k} e(j) \prod_{i=1}^k h(x_i, y_i, c_i) \prod_{i=k+1}^p x_i h(x_i - 1, y_i, c_i)$$

where $D^{(1)} = \text{diag}(h(x_1, y_1, c_1), h(x_2, y_2, c_2), \dots, h(x_p, y_p, c_p))$

and $D_h^{(2)} = \text{diag}(x_1 h(x_1 - 1, y_1, c_1), x_2 h(x_2 - 1, y_2, c_2), \dots, x_p h(x_p - 1, y_p, c_p))$.

Hence, we get

$$(1.25) \quad a(x, m) = \left[\prod_{i=1}^p \left\{ \frac{x_i^{y(i) + c_i + d_{ii}}}{x_i! y(i)!} \right\} \right] |D_y - D_x D|$$

where $y(i) = y_i - d_{ii} c_i$ for $i = 1, 2, \dots, p$,

$D_y = \text{diag}(y_1, \dots, y_p)$, $D_x = \text{diag}(x_1, \dots, x_p)$ and $y = m + D'x$.

Thus, the multivariate extensions of (1.14) and (1.16) are given by (1.7) and (1.8) respectively where $a(x, m)$ is defined by (1.25). We shall call (1.8) with (1.25) the multivariate Lagrangian Markov-Polya distribution, and (1.7) with (1.25) will tend to the multivariate Lagrangian Poisson as $c_i \rightarrow 0$ for all $i = 1, 2, \dots, p$. The distribution (1.7) with (1.25) was denoted by Khatri (1983 b) the multivariate Lagrangian negative binomial distribution, but it can contain many other distributions as special cases.

The Lagrangian multinomial distribution will be given by (1.1)

when $\exp(Q(\theta)) = (1 - \sum_j \theta_j)^{-a_0}$ and $\phi_i = \ln \theta_i - a_i \ln(1 - \sum_{j=1}^p \theta_j)$

for $i = 1, 2, \dots, p$.

The coefficient $a(x)$ is given by

$$(1.26) \quad a(x) = \frac{a_0}{a_0 + \sum_{i=1}^p a_i x_i} \left(a_0 + \sum_{i=1}^p a_i x_i + \sum_{i=1}^p x_i \right)$$

with

$$(1.27) \quad \binom{y}{x_1, x_2, \dots, x_p} = \frac{y!}{\left(\prod_{i=1}^p x_i!\right) \left(y - \sum_{i=1}^p x_i\right)!} \quad \text{for } y \geq \sum_{i=1}^p x_i, \\ = 0 \text{ otherwise}$$

Using these coefficients, we can obtain the quasi-Polya distributions for survival distribution as

$$(1.28) \quad s(x|z) = P(X=x|Z=z) = \frac{\left\{ \prod_{i=1}^p \binom{z_i}{x_i} \right\} q(a_0, a, x) q(b_0, a, z-x)}{q(a_0 + b_0, a, z)}$$

$$\text{where } q(a_0, a, x) = a_0 \left(a_0 + \sum_{i=1}^p a_i x_i + 1 \right)^{\left(\sum_{i=1}^p x_i - 1 \right)} \quad \text{and}$$

$$m(y, 1) = m(y) = m(m+1) \dots (m+y-1).$$

For the above two distributions, one can refer to Khatri (1983a) for details.

1.B Rao-Rubin conditions and characterization

For $p=1$, Rao and Rubin (1964) characterized $h(z)$ when the survival distribution is binomial with parameter π (fixed between 0 and 1) using the RR condition which is given by

$$(1.29) \quad P(X=x) = P(X=x|\text{undamaged}) = P(X=x|\text{damaged}) \text{ for all } x.$$

Recently, Rao et al. (1980) characterized $h(z)$ when the survival distribution is binomial and

$$(1.30) \quad P(X=x|\pi_1) = P(X=x|\text{undamaged}, \pi_2)$$

for all x and for fixed $0 < \pi_1, \pi_2 < 1$. The condition for

fixed $0 < \pi_1, \pi_2 < 1$ has been extended to two sets of values $\pi_1 < \pi_2, \pi'_1 < \pi'_2$ such that $(\log \pi_1 - \log \pi_2)/(\log \pi'_1 - \log \pi'_2)$ is irrational and this situation characterizes the negative binomial distribution of $h(z)$.

The above condition (1.29) is known as the RR condition. This condition was extended by Talwalker (1970) for $p = 2$ to:

$$\begin{aligned} (1.31) \quad P(X=x) &= P(X=x|\text{undamaged}) = P(X=x|z_1=x_1, z_2 > x_2) \\ &= P(X=x|z_1 > x_1, z_2 = x_2) \\ &= P(X=x|z_1 > x_1, z_2 > x_2) \quad \text{for all } x \end{aligned}$$

and she characterized the distribution of Z when the survival distribution is double binomial with parameters π_1 and π_2 using (1.31). Srivastava and Srivastava (1970), Aczel (1972) and Patil and Ratnaparkhi (1975) established the result of Talwalker (1970) using the condition:

$$(1.32) \quad P(X=x) = P(X=x|\text{undamaged}) \quad \text{for all } x.$$

This will be termed as RR condition for the multivariate situation. This means that the observed distribution under the condition of no damage is the same as the unobserved distribution. Such situations generally arise in socio-economic data. Further, we observe that Aczel (1972) and Patil and Ratnaparkhi (1975) in the proof of their results used the above condition (1.32) for all $0 < \pi_1, \pi_2 < 1$. Shanbhag (1974, 1977) showed that the result is not valid if the condition on the parameters is omitted. For the bivariate situation, Shanbhag (1977) considered the RR condition as

$$(1.33) \quad P(X=x) = P(X=x|Z=x) = P(X=x|z_1=x_1, z_2 > x_2)$$

for all x and characterized Talwalker's result (1970) using the double binomial distribution as the survival distribution. The purpose of this paper is to extend the above results to the multivariate situation using the condition (1.32) or

$$(1.34) \quad P(X=x) = P(X=x | Z=x) = P(X=x | Z=x+e_i)$$

for all x and for $i=1,2,\dots,p-1$ where e_i is the i -th column vector of the identity matrix I_p . In using the condition (1.32) for $p \geq 2$, we have to use its validity for $\theta \in D$ where θ is a parameter vector of the survival distribution, and this is similar to the situations considered by Aczel (1972) and Patil and Ratnaparkhi (1975) for $p=2$. The problem of using the condition (1.30), namely,

$$P(X=x | \theta_1) = P(X=x | \text{undamaged}, \theta_2) \quad \text{for all } x \text{ and for some } \theta_1, \theta_2 \in D,$$

for the characterization of $h(z)$ will be considered in a later communication. Notice that (1.34) is weaker than the condition (1.33) and hence this will generalize the results of Shanbhag (1977) for $p \geq 2$.

The main results on the characterization are given in the next section.

2. CHARACTERIZATION BY THE ADDITIVE DAMAGE MODEL

Shanbhag (1977) has assumed the survival distribution

$$(2.1) \quad s(x|z) = P(X=x | Z=z) = \alpha(x) \cdot \beta(z-x)/a(z)$$

for $x_i = 0,1,\dots,z_i$ and $i=1,2,\dots,p$ where $\beta(e_i) > 0$ for all $i=0,1,2,\dots,p$.

He considered only the situation $p = 2$ and established the distribution of Z as

$$(2.2) \quad c a(z) \exp(m'z),$$

c being a constant depending on m , under the condition (1.33). Obviously, if (2.1) and (2.2) hold, then X and Y are independently distributed and hence (1.33) holds. We give the result below for any p :

Theorem 1. Assume that the survival distribution is given by (2.1). Then, the distribution of Z is given by

$$(2.3) \quad P(Z = z) = c(z_{r+1}, \dots, z_p) a(z) \exp\left(\sum_{i=1}^r m_i z_i\right)$$

for some $m_1, \dots, m_r$, where $c(z_{r+1}, \dots, z_p)$ depends on $z_{r+1}, \dots, z_p$ iff

$$(2.4) \quad P(X = z | Z = x) = P(X = x | Z = x + e_i), \quad i = 1, 2, \dots, r \quad (\leq p),$$

for all x and $\beta(y) > 0$, $a(y) > 0$ for all $y \in T$.

Proof: Let the distribution of Z be given by $h(z)$ and let $f(z) = h(z)/a(z)$. Then, the joint density of X and Y is given by

$$f(x+y) = \alpha(x)\beta(y) \quad \text{for all } x \text{ and } y \in T.$$

$$\text{Let } g(e) = \sum_{x \in T} f(x+e)\alpha(x) \quad \text{and} \quad g(0) = \sum_{x \in T} f(x)\alpha(x).$$

Then, the given condition (2.4) is equivalent to

$$f(x)\alpha(x)/g(0) = f(x+e)\alpha(x)/g(e_i) \quad \text{for } i = 1, 2, \dots, r.$$

Taking $g(e_i)/g(0) = \exp(m_i)$ for $i = 1, 2, \dots, r$, we get

$$(2.5) \quad f(x+e_i) = f(x)\exp(m_i) \quad \text{for all } x \in T \text{ provided } \alpha(x) \neq 0.$$

Suppose for some $d \in T$, $f(d) = 0$. Then (2.5) shows that $f(d+e_i) = 0$ for $i = 1, 2, \dots, r$ and hence $f(x) = 0$ for all $x_i \geq d_i$, $i = 1, 2, \dots, r$, $x_i = d_i$, $i = r+1, \dots, p$. Hence, it is easy to see that the solution to (2.5) is

$$(2.6) \quad f(x) = c(x_{r+1}, \dots, x_p) \exp\left(\sum_{i=1}^r m_i x_i\right)$$

where $c(x_{r+1}, \dots, x_p)$ is a function of $x_{r+1}, \dots, x_p$.

(2.6) establishes (2.3) under (2.4). The converse is immediate.

Remark 1. When $p = r$, then (2.3) is equivalent to (2.2). The condition (2.4) for $r = p$ is different from (1.34). Suppose (1.34) holds. Then, by Theorem 1, using $p-1$ conditions, we get

$$P(Z=z) = c(z_p)a(z)\exp\left(\sum_{i=1}^{p-1} m_i z_i\right)$$

for some $m_1, \dots, m_{p-1}$ and $c(z_p)$ depends on z_p only. Then, we have to consider the condition

$$P(X=x) = P(X=x|Z=x) \quad \text{for all } x,$$

which is equivalent to

$$\sum_{y \in T} c(x_p + y_p) \beta(y) \exp\left(\sum_{i=1}^{p-1} m_i y_i\right) = c(x_p)/g(0) \quad \text{for all } x_p.$$

This is a univariate problem and its solution for $c(x_p)$ is given by $(\text{constant}) \cdot \exp(m_p x_p)$. Using this we get (2.2) if (2.6) is given (see Shanbhag (1977, Lemma-on p. 642)). The converse is immediate. Thus, we get

Corollary 1. Let the survival distribution be given by (2.1). Then, the distribution of Z is given by (2.2) iff (1.34) holds.

Remark 2. For $p=2$, Shanbhag, (1974, 1977) pointed out that the characterization based on (1.32) cannot determine (2.2) unless some conditions on the parameters of the survival distribution are imposed.

Theorem 2. Let the survival distribution be given by (1.8) for all $m, n \in \mathcal{D}_1$ for a fixed point $w = m+n \in \mathcal{D}_1$. If the distribution Z does not depend on m , then the distribution of Z is given by (1.7) with $m = w$ iff (1.32) holds.

Proof. Let the distribution of Z be given by $h(z)$. Then, the joint probability distribution of X and Y is

$$h(x+y)a(x,m)a(y,n)/a(x+y, m+n)$$

for all x and $y \in T$ and $m, n \in \mathcal{D}_1$ for a fixed point $w = m+n \in \mathcal{D}_1$. Let us denote

$$(2.7) \quad G(m) = \sum_{x \in T} h(x)a(x,m)/a(x,w)$$

where m and n take any values in $\mathcal{D}_1$ and $m+n = w$.

Then, the condition (1.32) gives

$$G(m) \sum_{y \in T} h(x+y)a(y,w-m)/a(x+y,w) = h(x)/a(x,w)$$

because $a(0,m) \neq 0$ for all $m \in \mathcal{D}_1$. Now, multiplying the above expression by $a(x,m_1)$ for any $m_1 \in \mathcal{D}_1$ and then summing over all x , we get (after some simplification)

$$G(m)G(w-m+m_1) = G(m_1) \quad \text{for all } m_1 \text{ and } m \in \mathcal{D}_1$$

This is the multivariate extension of Cauchy's functional equation with $G(w) = 1$ and hence its general solution is

$$G(m) = c \exp(m' \theta) \quad \text{for some } \theta \in D \text{ and for all } m \in \mathcal{D}_1$$

where c is a constant so that $G(w) = 1$. This gives

$$(2.8) \quad G(m) = \exp((m-w)\theta) \quad \text{for some } \theta \in \mathcal{D}.$$

Now, (2.7) shows that $h(x)/a(x,w)$ is a coefficient of $a(x,m)$ in the expansion of $G(m)$ given by (2.8). Thus, we get (1.7) with $m=w$ as the distribution of Z . The converse is immediate. This proves Theorem 2.

Theorem 3. Let the distribution of Z be given by (1.7) for some $m \in \mathcal{D}_1$ and for all $\theta \in \mathcal{D}$, and let the RR condition (1.32) hold for all $\theta \in \mathcal{D}$. If the survival distribution does not depend on θ , then it is given by (1.8) after replacing $m+n$ by m and by $m-n$ so that $m-n, m, n \in \mathcal{D}_1$.

Proof: Let $s(x|z)$ be the survival distribution of X given at z . Then, the condition (1.32) gives

$$(2.9) \quad \sum_{z \in T_x} a(z,m) \exp(-m\theta) s(x|z) \theta^z / (f(\theta))^z \\ = a(x,m) s(x|x) \theta^x / (f(\theta))^x G_1(\theta)$$

where $T_x = \{z; z_i = x_i, x_i + 1, \dots, \text{ for all } i = 1, 2, \dots, p\}$ and

$$(2.10) \quad G_1(\theta) = \sum_{x \in T} a(x,m) s(x|x) \theta^x / (f(\theta))^x.$$

Let us denote

$$(2.11) \quad G_2(\theta) = \sum_{z \in T} a(z,m) s(0|z) \theta^z / (f(\theta))^z a(0,m).$$

Now, taking $x=0$ in (2.9) and $a(0,m) \neq 0$, $s(0|0) = 1$, we get

$$G_2(\theta) \exp(-\theta' m) = \{G_1(\theta)\}^{-1} \quad \text{for all } \theta \in \mathcal{D}$$

or

$$(2.12) \quad \exp(\theta'm - m'e) = \exp(-m'e)G_1(\theta)G_2(\theta) \quad \text{for all } \theta \in \mathcal{D}$$

$$\text{where } e' = (1, 1, \dots, 1) = \sum_{i=1}^p e_i'$$

$$\text{Taking } g_1(\theta) = G_1(\theta)\exp(-m'e + m'\alpha)$$

$$\text{and } g_2(\theta) = G_2(\theta)\exp(-m'\alpha) \quad \text{for some } \alpha \text{ such that}$$

$$0 \leq \alpha_i \leq 1, \quad i=1, 2, \dots, p. \quad (2.12) \text{ can be rewritten as } (2.13) \\ \exp(\theta'm - m'e) = g_1(\theta)g_2(\theta).$$

Notice that $\exp(\theta'm - m'e)$ is a probability generating function of a Poisson distribution with parameters m , and the components of a Poisson distribution are Poisson distributions (see for example, Shalaysky (1959) or Lukacs and Laha (1964, p. 160)). Hence, we must have

$$(2.14) \quad G_1(\theta) = \exp((m-n)'\theta) \quad \text{and} \quad G(\theta) = \exp(n'\theta)$$

where $n' = (m_1\alpha_1, m_2\alpha_2, \dots, m_p\alpha_p)$, $n \in \mathcal{D}_1$, $m \in \mathcal{D}_1$ and $(m-n) \in \mathcal{D}_1$. (2.10) and (2.14) indicate that $a(x, m)s(x|x)$ is a coefficient of $\theta^x/(f(\theta))^x$ in the expansion of $\exp((m-n)'\theta)$ and using (1.7) we get

$$(2.15) \quad s(x|x) = a(x, m-n)/a(x, m).$$

Using (2.15) in (2.9), we get

$$\sum_{z \in \mathcal{I}_x} a(z, m)\exp(-m'\theta)s(x|z)\theta^z/(f(\theta))^z \\ = a(x, m-n)\exp(-m'\theta + n'\theta)\theta^x/(f(\theta))^x$$

and the right-hand side of the above expression can be written as

$$\sum_{y \in T} a(z, m-n) a(y, n) \exp(-m'\theta) \theta^{x+y} / (f(\theta))^{x+y}$$

$$= \sum_{z \in T_x} a(x, m-n) a(z-x, n) \exp(-m'\theta) \theta^z / (f(\theta))^z.$$

Hence, we get

$$\sum_{z \in T_x} \{s(x|z) a(z, m) - a(x, m-n) a(z-x, n)\} \theta^z / (f(\theta))^z = 0$$

for all $\theta \in D$. This gives the required result for

$$s(x|z) = a(x, m-n) a(z-x, n) / (z, m).$$

Remark 3. Using (1.25) for $a(x, m)$, we get the generalized result which includes all the known results and some further new survival distributions. Consul (1975) established the result only for the Lagrangian univariate Poisson using the survival distribution as Lagrangian binomial. Patil and Ratnaparkhi (1975) took the hypergeometric distribution. Shanbhag's result (1974) in the case of Talwalker's situation (1970) by taking the double binomial survival distribution for $p = 2$ is a particular case of Theorems 2 and 3. Patil and Ratnaparkhi (1975) claim that they have proved the result for $p = 2$ by taking a double hypergeometric survival distribution but the proof seems to be incorrect. Our Theorems 2 and 3 give the general result.

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